CAMPUSCARE

A unified campus management platform

Full-Stack MERN Semester Project | CS1237

The Problem

Campus information is scattered across paper notices, chats and separate processes.

CampusCare brings Lost & Found, campus issue reporting, class communities and examination information into one role-based platform.

Users & Roles

Students • Teachers • Class Representatives • Admin

Students submit and track items and reports, join class groups and view exam information. Teachers manage assigned work and duties. Admin oversees users, reports, communities and exams.

Core MERN Architecture

React + Vite → Express REST API → MongoDB Atlas

React Router SPA and Context API • Express routes/controllers/middleware • Mongoose related collections • JWT authentication and role checks • Vercel + Render deployment targets

Database Design

Related collections, not a single-table CRUD app

Users, LostFound, LostFoundClaims, Reports, Communities, Exams, ExamSeats, ExamDuties, ExamRequests and ContactMessages. See docs/database-erd.md.

Advanced Feature 1: Rate Limiting

Protect sensitive endpoints from repeated requests

Login and public contact routes use express-rate-limit. Demo: exceed the configured request threshold and show HTTP 429 with a clear response.

Advanced Feature 2: Real-Time Communication

Community updates without manual refresh

Socket.IO authenticates the connection using the session token. Community room joins are checked against membership. A new post emits a community-post event to members.

Additional Integrations & Demo

GridFS • TTL Cache • Role-based flows

GridFS stores Lost & Found images. Community listings show cache HIT/MISS and cache invalidation. Demo student, teacher and admin workflows; verify live URLs and test credentials before presentation.